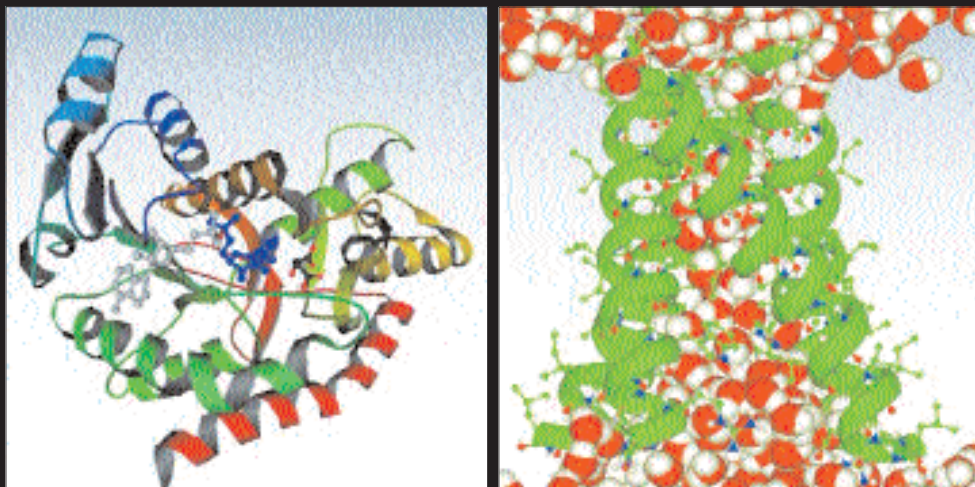


Catalytic Simulation



The image above (left) shows the enzyme malate dehydrogenase with reactants for catalytic process in place. The position where these reactants are bound is specifically tailored in shape and affinity to enormously accelerate this specific reaction and to be inert to other processes. The image on right is the M2 proton channel of the influenza virus. This ion channel plays an essential role in causing infection and is the site of attack of a class of channel-blocking flu drugs. All these are supercomputer simulations of the analysis done in the biomolecular segment

formulation and excessive processing, in terms of human hours. Before the advent of supercomputers wind tunnels were the only way of testing the aerodynamics of new aircraft designs. Even today there are certain limitations to simulations done on supercomputers, and resorting to wind tunnels to verify the design for a new aeroplane seems a more appropriate option. But in due course powerful supercomputers will be able to overcome the existing obstacles.

Weather prediction is another area where supercomputers are employed since the analyses here require massive calculations and prediction schemes, which the machine does by performing millions of operations in a second. Using computer predictions you can draw conclusions about the pressure that might build up in a particular area, the possibility of rainfall or storms. Forecasters and weather scientists use computers that can crank through about 10 million calculations per second. Jet, one of the fastest supercomputers used for weather prediction, can perform 300 million calculations per second which would rise up to 5 trillion calculations per second as engineers try to add more

processing power to the system. What really makes this \$17 million, 273-processor supercomputer different from most other supercomputers is its powerful Linux architecture.

Where is supercomputing headed?

Silicon-based chips are reaching their limits of miniaturisation and speed. But chips could get smaller and faster still. We have better material inside the human body, which can act as the raw material for building a powerful supercomputer. DNA has the capability to process and store data way beyond the capabilities of today's fastest supercomputers. The biochip is in its infancy right now, but that is where the future of supercomputing lies. Raw materials required to manufacture bio-computers will never get depleted as long as living organisms exist on earth. Besides, unlike the raw materials that go into the production of today's computers, these are not toxic. We can hope that with the computational power and storage facilities that DNA-based computers will bring with them, we can continue on our quest for a better life and for solutions for the mysteries of the universe. 

What's parallel processing all about?

While speaking of supercomputers, we frequently talk about parallel processing. So what's parallel processing anyway? To put it simply, it's processing more than one datum at a time. When we have multiple processors in hand, we like the data to be shared between all the processors to quicken the speed of computation. But they have to talk to each other with the datum that each one of them possesses. Hence they talk to a common mediator called the memory. This essentially describes the parallel computation process, in which a computer handles more than one instruction per second and performs computation at extremely high speeds.

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